AI PORTFOLIO PROJECT | NEWS & MEDIA

Spectrum

See every news story from every side, mapped on an interactive bias spectrum.

The problem

Algorithms trap people in filter bubbles. The same event is told completely differently by different outlets, and nobody has time to read ten versions.

The concept

Spectrum clusters articles about the same event from hundreds of sources. Each story becomes a card that you can expand into a spectrum: outlets plotted from left to right and by factual reliability. The AI writes a neutral summary, highlights which facts every side agrees on, which facts only one side mentions, and where wording differs ('protesters' vs 'rioters').

Core features

- Automatic event clustering across many RSS feeds
- Neutral AI summary with citations to each source
- 'Agreed facts vs disputed claims' breakdown
- Loaded-language detector highlighting framing differences
- Personal 'bubble meter' showing how balanced your reading is

Signature UI / UX moments

- Horizontal spectrum with outlet logos that slide into place when a story opens
- Side-by-side headline comparison with highlighted word differences
- Editorial, newspaper-inspired typography with a modern dark mode

Tech stack

Next.js + D3.js
Cron-driven ingestion (Vercel Cron / Queues) from RSS feeds
Embeddings + clustering (HDBSCAN or agglomerative) in a Python service
AI SDK for summaries and claim extraction
Neon Postgres + pgvector

How the AI works

A scheduled pipeline ingests feeds, embeds articles and clusters them into events. For each event, an LLM extracts atomic claims per article, then claims are matched across sources to compute agreement. Every summary sentence links back to its sources.

4-week MVP roadmap

Week 1	RSS ingestion from 30 outlets, embeddings, clustering
Week 2	Neutral summaries with citations
Week 3	Claim extraction and agreement matrix, spectrum UI
Week 4	Loaded-language highlights, bubble meter, polish

Stretch goals

- Browser extension that shows the spectrum on any news site
- Daily audio briefing
- Historical tracking of how a story's framing evolves

Why it impresses hiring managers
A full data pipeline (ingestion, clustering, extraction, evaluation) behind a polished UI. Shows you can build systems, not just interfaces, and the topic sparks conversation in interviews.

30-second demo script
Open today's top story. Twenty outlets slide onto the spectrum. Toggle 'compare headlines' and see the same event worded five different ways, with the agreed facts at the top.